

Dyadic-Comb Moments of the Fabius and Rvachev Functions

Exact rational formulae, sinc-zero superconvergence, spectral Dirichlet values, and iterated sums

Research report prepared for Vladimir Reshetnikov

28 August 2026

Abstract

Let F be the Fabius function on $[0, 1]$, let $\mathfrak{u}$ be Rvachev's compactly supported bump, and put $N = 2^m$. This report studies the weighted dyadic-comb rule

$$S_{m,p} = \frac{1}{N} \sum_{k=0}^N \left(\frac{k}{N}\right)^p F\left(\frac{k}{N}\right)$$

and its extensions to complex powers and repeated summation. The first main result is a finite Thue–Morse/Appell formula for every dyadic value and hence a closed rational expression for $S_{m,p}$ at arbitrary $m, p \in \mathbb{N}_0$. Its generating function is rational after negative-index polylogarithms are expanded, and an $O(m2^m)$ sparse-difference algorithm replaces the apparent quadratic convolution.

The second main result explains an unexpectedly strong phenomenon. The Fourier transform

$$H(z) = \widehat{\mathfrak{u}}(z) = \prod_{\nu \geq 1} \operatorname{sinc}\left(\frac{z}{2^\nu}\right)$$

has a zero of multiplicity $m + 1 + v_2(\ell)$ at every nonzero alias $2\pi 2^m \ell$. Consequently the full Rvachev comb integrates every polynomial of degree at most m exactly. For the Fabius half-bump this gives, for $m \geq p + 1$,

$$S_{m,p} = I_p + \frac{1}{2N} + \sum_{q=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2q}}{(2q)!} p^{2q-1} N^{-2q}, \quad I_p = \int_0^1 x^p F(x) \, dx,$$

with no unshown remainder. The boundary case $m = p$ is also exact when p is even. A first-alias calculation produces a new family of exact spectral Dirichlet evaluations; for example

$$\sum_{\substack{q \geq 1 \\ q \text{ odd}}} \frac{H(\pi q)}{q^2} = \frac{\pi^2}{18}, \quad \sum_{\substack{q \geq 1 \\ q \text{ odd}}} \frac{H(\pi q)}{q^4} = \frac{23\pi^4}{4050}.$$

For arbitrary $\alpha \in \mathbb{C}$, both the continuous Mellin moment and the finite comb are entire functions of α . Integer truncation is replaced by a universal Euler–Maclaurin expansion with falling factorials. Finally, an inclusive n -fold sum is reduced exactly to n ordinary comb moments by a binomial kernel; its continuum limit is a beta-weighted Fabius moment. The report includes exact checks, high-precision experiments, conjectures about alias sharpness and spectral positivity, and a recursive inventory of the repository corpus used to avoid duplicating its existing integral, asymptotic, interpolation, and q -series developments.

Contents

1 Scope, corpus audit, and novelty boundary

2

2	Normalization and basic identities	2
2.1	Rvachev’s bump and the Fabius half-bump	2
2.2	A probabilistic model	3
3	Continuous moments in closed form	3
4	A finite Thue–Morse formula for every dyadic value	4
5	Closed forms for polynomially weighted dyadic combs	5
5.1	A direct finite formula	5
5.2	Generating function and an $O(m2^m)$ algorithm	5
6	Poisson summation and dyadic exactness	6
6.1	Exact polynomial quadrature for the full Rvachev bump	6
6.2	The one-sided Fourier transform	7
6.3	An exact finite Euler–Maclaurin formula	7
6.4	All levels: an exact alias remainder	8
7	The first alias defect and spectral Dirichlet values	9
8	Other comb conventions and Rvachev/Fabius transforms	10
9	Negative, fractional, and complex powers	10
9.1	Entire Mellin dependence	10
9.2	Universal endpoint expansion	11
10	Iterated sums of arbitrary integer order	12
10.1	Fractional-order repeated sums	13
11	Exact and numerical checks	13
12	Further deductions	13
12.1	A quadrature-design interpretation	13
12.2	Two-adic filtration of higher defects	13
12.3	Richardson extrapolation	14
13	Conjectures	14
14	Promising research directions	14
15	Conclusion	15
A	Recursive repository inventory	15
B	Reproducibility notes	15

1 Scope, corpus audit, and novelty boundary

The source corpus was taken recursively from

[https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/
FabiusFunction/docs](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs).

The checked snapshot contains [not shipped] `.tex` files at Git commit [not shipped]; the combined content digest begins [not shipped]. The machine-readable audit is included as `corpus_audit.txt`, and the full relative-path inventory appears in [section A](#).

The corpus already develops a substantial common foundation: exact dyadic values, Thue–Morse structure, q -Pochhammer and q -binomial forms, sinc products and Fourier transforms, continuous moments, Bell/Bernoulli polynomials, endpoint asymptotics involving Lambert W , inverse-Fabius questions, interpolation, Richardson extrapolation, and several classes of integrals. Those ingredients are used rather than re-presented as purported new discoveries. The emphasis here is deliberately narrower and discrete:

- (i) a closed generating object for *weighted comb sums*, not merely a formula for an individual dyadic value;
- (ii) exactness and defect formulae obtained from the *multiplicity* of dyadic sinc zeros under Poisson summation;
- (iii) spectral Dirichlet values extracted from the first nonzero alias;
- (iv) complex powers, including negative powers, treated through an entire Mellin parameter; and
- (v) inclusive, strict, and fractional-order iterated sums.

A repository audit can rule out literal and close structural duplication in a finite corpus; it cannot certify priority against all unpublished mathematics. Accordingly, statements proved below are labeled theorems, computational observations are labeled as such, and wider arithmetic patterns are explicitly presented as conjectures.

2 Normalization and basic identities

2.1 Rvachev’s bump and the Fabius half-bump

Throughout,

$$\operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0, \end{cases} \quad H(z) = \prod_{\nu=1}^{\infty} \operatorname{sinc}\left(\frac{z}{2^{\nu}}\right).$$

With the Fourier convention $\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-izx} dx$, Rvachev’s function $u = \operatorname{up}$ is the even C^{∞} function supported on $[-1, 1]$ satisfying $\widehat{u} = H$, $u(0) = 1$, and $\int u = 1$. On $[0, 1]$ we use

$$F(x) = u(x-1) = u(1-x). \tag{1}$$

Thus

$$F(x) + F(1-x) = 1, \quad u(x) + u(1-x) = 1 \quad (0 \leq x \leq 1). \tag{2}$$

Both F at 0 and $1 - F$ at 1 are flat: every one-sided derivative vanishes. Equivalently,

$$F(x) = O(x^A), \quad 1 - F(1-x) = O(x^A) \quad (x \downarrow 0) \tag{3}$$

for every $A > 0$. This flatness is what makes negative powers harmless later.

2.2 A probabilistic model

Let $U_1, U_2, \dots$ be independent uniform random variables on $[-1, 1]$, and put

$$X = \sum_{\nu=1}^{\infty} 2^{-\nu} U_{\nu}. \quad (4)$$

Then X has density u , because its characteristic function is H . Its moment-generating function is

$$M(t) = \mathbb{E}e^{tX} = \prod_{\nu=1}^{\infty} \frac{\sinh(t/2^{\nu})}{t/2^{\nu}}. \quad (5)$$

Write $\mu_r = \mathbb{E}X^r$. Symmetry gives $\mu_{2r+1} = 0$. From

$$\log \frac{\sinh z}{z} = \sum_{r \geq 1} \frac{2^{2r-1} B_{2r}}{r(2r)!} z^{2r}$$

we obtain the even cumulants

$$\kappa_{2r} = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0, \quad (6)$$

and hence

$$\mu_n = Y_n(0, \kappa_2, 0, \kappa_4, \dots, \kappa_n), \quad (7)$$

where Y_n is the complete exponential Bell polynomial. In particular all μ_n are rational. The first values are $\mu_0 = 1$, $\mu_2 = 1/9$, and $\mu_4 = 19/675$.

Define the Appell moment polynomials

$$P_m(y) = \mathbb{E}(y - X)^m = \sum_{r=0}^m (-1)^r \binom{m}{r} \mu_r y^{m-r} = m! [t^m] e^{yt} M(t). \quad (8)$$

Only the even r survive, so the signs disappear in the explicit sum.

3 Continuous moments in closed form

The continuous moments are needed both as limiting integrals and as constants in the exact comb formula.

Theorem 3.1 (Exponential generating function). *Let $I_p = \int_0^1 x^p F(x) dx$. Then*

$$\sum_{p=0}^{\infty} I_p \frac{t^p}{p!} = \frac{e^t}{t} - \frac{e^t M(t)}{e^t - 1}. \quad (9)$$

Consequently

$$I_p = \frac{1}{p+1} \left[1 - \sum_{j=0}^{p+1} \binom{p+1}{j} \mu_j B_{p+1-j}(1) \right], \quad (10)$$

where $B_n(x)$ is the Bernoulli polynomial. In particular, $I_p \in \mathbb{Q}$ for every $p \in \mathbb{N}_0$.

Proof. Set $A(t) = \int_0^1 u(x) e^{tx} dx$. Evenness gives $A(t) + A(-t) = M(t)$, while the partition identity (2) gives

$$A(t) + e^t A(-t) = \frac{e^t - 1}{t}.$$

Solving the two equations yields

$$A(-t) = \frac{1}{t} - \frac{M(t)}{e^t - 1}.$$

By (1),

$$\int_0^1 F(x)e^{tx} dx = e^t A(-t),$$

which proves (9). Expanding $te^t/(e^t - 1) = \sum_{n \geq 0} B_n(1)t^n/n!$ and using $M(t) = \sum \mu_j t^j/j!$ gives (10). $\square$

The same calculation gives the positive-half moments of u :

$$a_r := \int_0^1 x^r u(x) dx = \frac{1}{r+1} \sum_{j=0}^{r+1} \binom{r+1}{j} \mu_j B_{r+1-j}(1) = \frac{1}{r+1} - I_r. \quad (11)$$

These formulas connect the Bell-polynomial moment algebra to the Bernoulli polynomials without invoking numerical quadrature.

4 A finite Thue–Morse formula for every dyadic value

Let

$$N = 2^m, \quad h = N^{-1}, \quad \tau(a) = (-1)^{s_2(a)},$$

where $s_2(a)$ is the binary digit sum.

Theorem 4.1 (Dyadic Appell–Thue–Morse formula). *For $m \in \mathbb{N}_0$ and $0 \leq k \leq N$,*

$$F\left(\frac{k}{N}\right) = \frac{2^{-m(m+1)/2}}{m!} \sum_{a=0}^{k-1} \tau(a) P_m(2(k-a)-1), \quad (12)$$

where the empty sum for $k=0$ is zero.

Proof. Put $n = m+1$ and split the random series (4) after n terms. After multiplication by 2^n ,

$$2^n X = \sum_{\nu=1}^n 2^{n-\nu} U_\nu + X',$$

where X' is an independent copy of X . The density of the finite sum on the right is a nonuniform Irwin–Hall spline. Since its half-widths are $1, 2, 4, \dots, 2^{n-1}$, their subset sums are exactly the integers $0, \dots, 2^n - 1$, and inclusion–exclusion gives

$$f_n(y) = \frac{2^{-n(n+1)/2}}{(n-1)!} \sum_{a=0}^{2^n-1} \tau(a) (y + 2^n - 1 - 2a)_+^{n-1}.$$

The density scaling identity is

$$2^{-n} u(y/2^n) = \mathbb{E} f_n(y - X').$$

At $x = k/N - 1$ one has $y = 2(k - N)$, and the truncated-power argument becomes $2(k-a) - 1 - X'$. Since $X' \in [-1, 1]$, it is nonnegative exactly when $a \leq k-1$ (endpoint equalities are immaterial). Taking the expectation produces $P_m(2(k-a)-1)$ and yields (12). $\square$

Example 4.2. For $m=2$, $P_2(y) = y^2 + 1/9$ and the prefactor is $1/16$. Thus

$$F(1/4) = \frac{1}{16} \left(1 + \frac{1}{9}\right) = \frac{5}{72}, \quad F(1/2) = \frac{1}{2}.$$

For $m=3$, $P_3(y) = y^3 + y/3$, and one obtains

$$(F(k/8))_{k=0}^8 = \left(0, \frac{1}{288}, \frac{5}{72}, \frac{73}{288}, \frac{1}{2}, \frac{215}{288}, \frac{67}{72}, \frac{287}{288}, 1\right).$$

5 Closed forms for polynomially weighted dyadic combs

5.1 A direct finite formula

For $p \in \mathbb{N}_0$ define the right-endpoint comb

$$S_{m,p} = h \sum_{k=0}^N x_k^p F(x_k), \quad x_k = kh. \quad (13)$$

The $k = 0$ term is zero, including at $p = 0$ under the natural convention. Substitution of (12) immediately gives the requested finite closed form.

Corollary 5.1 (Finite rational comb formula). *For every $m, p \in \mathbb{N}_0$,*

$$S_{m,p} = \frac{2^{-m(m+1)/2}}{m!N^{p+1}} \sum_{k=1}^N k^p \sum_{a=0}^{k-1} \tau(a) P_m(2(k-a)-1). \quad (14)$$

In particular $S_{m,p} \in \mathbb{Q}$.

After putting $d = k - a$, the inner power sums can be written with Bernoulli polynomials:

$$\sum_{d=1}^L d^r = \frac{B_{r+1}(L+1) - B_{r+1}(1)}{r+1}.$$

Expanding $(a+d)^p P_m(2d-1)$ therefore turns (14) into a single finite Thue–Morse sum of explicit Bernoulli polynomials. This is useful for symbolic elimination; the generating function below is more compact.

5.2 Generating function and an $O(m2^m)$ algorithm

Define

$$\Theta_m(z) = \sum_{a=0}^{N-1} \tau(a) z^a = \prod_{j=0}^{m-1} (1 - z^{2^j}), \quad (15)$$

$$B_m^*(z) = \sum_{d \geq 1} P_m(2d-1) z^d, \quad (16)$$

$$G_m(z) = \frac{2^{-m(m+1)/2}}{m!} \Theta_m(z) B_m^*(z). \quad (17)$$

Theorem 5.2 (Dyadic-value generating function). *The coefficient of z^k in $G_m(z)$ equals $F(k/N)$ for $0 \leq k \leq N$ and equals 1 for every $k \geq N$. Hence*

$$A_m(z) := G_m(z) - \frac{z^{N+1}}{1-z} = \sum_{k=0}^N F(k/N) z^k \quad (18)$$

is a polynomial, and

$$S_{m,p} = N^{-p-1} \left(z \frac{d}{dz} \right)^p A_m(z) \Big|_{z=1}. \quad (19)$$

All apparent singularities at $z = 1$ in this representation are removable.

Proof. The coefficient convolution in (17) is precisely (12). Once $k \geq N$, all $a = 0, \dots, N-1$ occur. The Prouhet cancellation

$$\sum_{a=0}^{N-1} \tau(a) a^r = 0 \quad (r < m), \quad \sum_{a=0}^{N-1} \tau(a) a^m = (-1)^m m! 2^{m(m-1)/2}$$

shows that only the leading term of the monic polynomial P_m survives, and its value after the factor 2^m from $P_m(2(k-a)-1)$ is exactly the reciprocal of the prefactor. Thus the tail coefficients are 1. Applying the Euler operator $z d/dz$ multiplies the k th coefficient by k , proving (19). $\square$

The series $B_m^\star$ is rational. Indeed,

$$B_m^\star(z) = \sum_{r=0}^{\lfloor m/2 \rfloor} \binom{m}{2r} \mu_{2r} \sum_{j=0}^{m-2r} \binom{m-2r}{j} 2^j (-1)^{m-2r-j} \text{Li}_{-j}(z), \quad (20)$$

where every negative-index polylogarithm is rational: $\text{Li}_{-j}(z) = z A_j(z)/(1-z)^{j+1}$ with A_j an Eulerian polynomial. Equations (18)–(20) are therefore a finite algebraic closed form for every natural p .

Computationally, one should not form the double sum. Initialize the coefficient array with $P_m(2d-1)$ and multiply successively by $(1-z), (1-z^2), \dots, (1-z^{2^{m-1}})$. Each multiplication is one sparse backward difference, so all $N+1$ dyadic values cost $O(mN)$ arithmetic operations and $O(N)$ storage. The accompanying `experiments.py` implements this exact algorithm with rational arithmetic.

6 Poisson summation and dyadic exactness

6.1 Exact polynomial quadrature for the full Rvachev bump

For a polynomial P define

$$Q_m[P] = h \sum_{k=-N}^N u(kh) P(kh).$$

Since u and all of its derivatives vanish at ± 1 , no endpoint convention is needed.

Theorem 6.1 (Full-bump dyadic exactness). *If $\deg P \leq m$, then*

$$Q_m[P] = \int_{-1}^1 u(x) P(x) dx. \quad (21)$$

Equivalently, for $0 \leq p \leq m$,

$$h \sum_{k=-N}^N (kh)^p u(kh) = \mu_p. \quad (22)$$

Proof. For a monomial, Poisson summation gives

$$h \sum_{k \in \mathbb{Z}} (kh)^p u(kh) = \sum_{\ell \in \mathbb{Z}} i^p H^{(p)}(2\pi N \ell).$$

At $z_\ell = 2\pi 2^m \ell \neq 0$, each factor $\text{sinc}(z/2^\nu)$ with $1 \leq \nu \leq m+1+v_2(\ell)$ vanishes simply. Thus

$$\text{ord}_{z=z_\ell} H(z) = m+1+v_2(\ell). \quad (23)$$

Every nonzero alias therefore vanishes after at most m derivatives. Only $\ell = 0$ remains, and its value is the moment μ_p . $\square$

This is stronger than rapid convergence: at each level the dyadic comb is an exact quadrature rule on a growing polynomial space. For even $p > 0$ it also gives the useful half-grid identity

$$h \sum_{k=0}^N (kh)^p u(kh) = \frac{\mu_p}{2} \quad (2 \mid p, 0 < p \leq m), \quad (24)$$

while for $p = 0$ the center sample contributes $h \sum_{k=0}^N u(kh) = (1 + h)/2$.

6.2 The one-sided Fourier transform

Let $g_0(x) = F(x)$ on $[0, 1]$ and zero outside, with the symmetric half-value at the jump $x = 1$. A key identity is

$$\widehat{g}_0(z) = -\frac{e^{-iz}}{iz} + \frac{H(z)}{e^{iz} - 1}. \quad (25)$$

To derive it, put $U_+(z) = \int_0^1 u(x) e^{-izx} dx$. Evenness and (2) give

$$U_+(z) + U_+(-z) = H(z), \quad U_+(z) + e^{-iz} U_+(-z) = \frac{1 - e^{-iz}}{iz}.$$

Solving and shifting by (1) yields (25). For $g_p(x) = x^p F(x)$,

$$\widehat{g}_p(z) = i^p \frac{d^p}{dz^p} \widehat{g}_0(z). \quad (26)$$

At a nonzero dyadic alias, the quotient

$$R(z) := \frac{H(z)}{e^{iz} - 1} \quad (27)$$

has zero order

$$\text{ord}_{z=2\pi N\ell} R(z) = m + v_2(\ell). \quad (28)$$

The denominator removes exactly one of the sinc zeros. This is the source of the finite rule below.

6.3 An exact finite Euler–Maclaurin formula

The right rule (13) differs from the symmetric trapezoidal convention by $h/2$, because $F(1) = 1$ and $F(0) = 0$.

Theorem 6.2 (Fabius dyadic superconvergence). *Let $m, p \in \mathbb{N}_0$, $N = 2^m$, and I_p be given by (10). If $m \geq p + 1$, then*

$$S_{m,p} = I_p + \frac{1}{2N} + \sum_{q=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2q}}{(2q)!} p^{2q-1} N^{-2q}. \quad (29)$$

The same formula holds in the boundary case $m = p$ whenever p is even.

Proof. Poisson summation, with the half-value at $x = 1$, gives

$$S_{m,p} - \frac{h}{2} = I_p + 2\Re \sum_{\ell \geq 1} \widehat{g}_p(2\pi N\ell).$$

By (28), the contribution from $R(z)$ in (25) vanishes whenever $p < m$. It remains to differentiate the endpoint term $-e^{-iz}/(iz)$. At $z = 2\pi N\ell$,

$$i^p \frac{d^p}{dz^p} \left(-\frac{e^{-iz}}{iz} \right) = \sum_{r=0}^p \binom{p}{r} r! i^{1-r} z^{-r-1}.$$

Pairing ℓ and $-\ell$ retains only $r = 2q - 1$. Euler's evaluation of $\zeta(2q)$ converts the result to

$$\frac{B_{2q}}{(2q)!} p^{2q-1} N^{-2q},$$

which proves (29). If $p = m$ is even, the leading derivative of R has odd Fourier parity and its paired real part vanishes. Alternatively, combine theorem 6.1 with (24) and $u(x) = 1 - F(x)$. $\square$

For example,

$$\begin{aligned} S_{m,0} &= \frac{1}{2} + \frac{1}{2N}, \\ S_{m,1} &= \frac{13}{36} + \frac{1}{2N} + \frac{1}{12N^2} \quad (m \geq 2), \\ S_{m,2} &= \frac{5}{18} + \frac{1}{2N} + \frac{1}{6N^2} \quad (m \geq 2), \\ S_{m,3} &= \frac{1207}{5400} + \frac{1}{2N} + \frac{1}{4N^2} - \frac{1}{120N^4} \quad (m \geq 4). \end{aligned}$$

The formulas are exact identities at the stated dyadic levels, not asymptotic expansions.

For even p there is a second compact form. From (24), for $0 < p \leq m$ even,

$$S_{m,p} = \frac{1}{N^{p+1}} \sum_{k=1}^N k^p - \frac{\mu_p}{2} = \frac{B_{p+1}(N+1) - B_{p+1}}{(p+1)N^{p+1}} - \frac{\mu_p}{2}. \quad (30)$$

This identity makes the rationality and the boundary extension $m = p$ entirely transparent.

6.4 All levels: an exact alias remainder

At levels below the stabilization threshold, the missing term is still explicit.

Proposition 6.3 (Alias remainder). *For arbitrary $m, p \in \mathbb{N}_0$,*

$$S_{m,p} = I_p + \frac{1}{2N} + \sum_{q=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2q}}{(2q)!} p^{2q-1} N^{-2q} + \mathcal{R}_{m,p}, \quad (31)$$

where the absolutely convergent alias series is

$$\mathcal{R}_{m,p} = 2\Re \sum_{\ell=1}^{\infty} i^p \frac{d^p}{dz^p} \left(\frac{H(z)}{e^{iz} - 1} \right) \Big|_{z=2\pi N\ell}. \quad (32)$$

Only valuation classes $v_2(\ell) \leq p - m$ can contribute.

The last observation follows immediately from (28). It converts an infinite alias set into finitely many 2-adic valuation strata, each still containing rapidly decaying odd multiples. This is a useful starting point for higher-defect formulae.

7 The first alias defect and spectral Dirichlet values

Let

$$D(s) = \sum_{\substack{q \geq 1 \\ q \text{ odd}}} \frac{H(\pi q)}{q^s}. \quad (33)$$

Because $H(x)$ decreases faster than every inverse power on the real axis, this Dirichlet series defines an entire function of s .

Fix an odd q and put $z_0 = 2\pi 2^m 2^r q$, so $s_0 = m + r + 1$ sinc factors vanish. Removing those factors gives the local expansion

$$H(z) = -\frac{H(\pi q)}{z_0^{s_0}} (z - z_0)^{s_0} + O((z - z_0)^{s_0+1}). \quad (34)$$

Since $e^{iz} - 1 = i(z - z_0) + O((z - z_0)^2)$,

$$R(z) = \frac{iH(\pi q)}{z_0^{s_0}} (z - z_0)^{s_0-1} + O((z - z_0)^{s_0}). \quad (35)$$

At derivative order $p = m$, only the odd aliases ($r = 0$) can occur.

Theorem 7.1 (First-defect identity). *For odd $m \geq 1$,*

$$\begin{aligned} S_{m,m} - I_m - \frac{1}{2N} - \sum_{q=1}^{(m+1)/2} \frac{B_{2q}}{(2q)!} m^{2q-1} N^{-2q} \\ = \frac{2m!(-1)^{(m+1)/2}}{(2\pi N)^{m+1}} D(m+1). \end{aligned} \quad (36)$$

Consequently

$$\boxed{D(2r)/\pi^{2r} \in \mathbb{Q} \quad (r \geq 1).} \quad (37)$$

Proof. Insert (35) into (32) with $p = m$. The factor $\Re(i^{m+1}) = (-1)^{(m+1)/2}$ gives (36). Every other term in that equation is rational by theorems 3.1 and 5.1; solving for $D(m+1)$ proves (37). $\square$

The first two evaluations are

$$D(2) = \frac{\pi^2}{18}, \quad D(4) = \frac{23\pi^4}{4050}. \quad (38)$$

The first follows from the level-1 defect $S_{1,1} - (I_1 + 1/4 + 1/48) = -1/144$. For $m = 3$ one obtains

$$S_{3,3} = \frac{42751}{147456}, \quad S_{3,3} - \left(I_3 + \frac{1}{16} + \frac{1}{256} - \frac{1}{491520} \right) = \frac{23}{22118400},$$

which gives the second identity. Further exact values generated from (36) appear in ??.

The identity

$$H(\pi q) = \frac{2(-1)^{(q-1)/2}}{\pi q} H(\pi q/2) \quad (q \text{ odd})$$

connects $D(s)$ to a twisted half-scale series. This suggests a hierarchy of functional relations combining odd residue classes, dyadic dilation, and Thue–Morse signs.

8 Other comb conventions and Rvachev/Fabius transforms

Let

$$\Pi_{m,p} = h \sum_{k=0}^N (kh)^p = \frac{B_{p+1}(N+1) - B_{p+1}(0)}{(p+1)N^{p+1}}.$$

The basic transformations require no new sampling:

$$h \sum_{k=0}^N (kh)^p (1 - F(kh)) = \Pi_{m,p} - S_{m,p}, \quad (39)$$

$$h \sum_{k=0}^N (kh)^p F(1 - kh) = \Pi_{m,p} - S_{m,p}. \quad (40)$$

Because $u(x) = 1 - F(x)$ on $[0, 1]$, these are also the positive-half Rvachev sums.

The closed right rule, trapezoidal rule, and open/left rule satisfy

$$T_{m,p} = S_{m,p} - \frac{h}{2}, \quad L_{m,p} = S_{m,p} - h. \quad (41)$$

For the midpoint comb

$$M_{m,p} = h \sum_{k=0}^{N-1} \left(\frac{k+1/2}{N} \right)^p F\left(\frac{k+1/2}{N} \right),$$

splitting the level- $(m+1)$ grid into even and odd nodes gives the exact identity

$$M_{m,p} = 2S_{m+1,p} - S_{m,p}. \quad (42)$$

Thus every standard dyadic rule inherits the closed forms above.

For the full bump, [theorem 6.1](#) is the strongest statement:

$$h \sum_{k=-N}^N P(kh)u(kh) = \mathbb{E}P(X) \quad (\deg P \leq m).$$

Translations and affine rescalings of u merely transform the polynomial and the mesh. Tensor products therefore give an exact cubature rule on all coordinatewise degrees at most m for products of Rvachev bumps.

9 Negative, fractional, and complex powers

9.1 Entire Mellin dependence

For $\alpha \in \mathbb{C}$ define

$$S_m(\alpha) = h \sum_{k=1}^N (kh)^\alpha F(kh), \quad I(\alpha) = \int_0^1 x^\alpha F(x) dx, \quad (43)$$

using the real logarithm on $(0, 1]$. At $x = 0$ the product is defined by its limit 0.

Theorem 9.1 (Entire power parameter). *Both $S_m(\alpha)$ and $I(\alpha)$ are entire functions of α . Moreover,*

$$S_m(\alpha) = \frac{2^{-m(m+1)/2}}{m!N^{\alpha+1}} \sum_{k=1}^N k^\alpha \sum_{a=0}^{k-1} \tau(a) P_m(2(k-a)-1) \quad (44)$$

is an exact finite expression for arbitrary complex, fractional, or negative α .

Proof. The finite sum is manifestly entire. For $I(\alpha)$, fix a compact set $K \subset \mathbb{C}$. By flatness (3), choose A larger than $-\min_{\alpha \in K} \Re \alpha$ plus any prescribed number of logarithmic derivatives. Then $x^\alpha (\log x)^r F(x)$ is uniformly integrable for $\alpha \in K$. Differentiation under the integral is valid to every order. The finite formula follows from [theorem 4.1](#). $\square$

The continuous entire function has a rapidly convergent Newton–binomial series.

Proposition 9.2 (Fractional moment series). *For every $\alpha \in \mathbb{C}$,*

$$I(\alpha) = \sum_{r=0}^{\infty} (-1)^r \binom{\alpha}{r} a_r, \quad (45)$$

where a_r is given explicitly by (11). The convergence is locally uniform in α .

Proof. Use $F(x) = u(1-x)$ and expand $(1-y)^\alpha$ on $0 \leq y < 1$. Since u is flat at $y = 1$, integration by parts or a beta estimate gives $a_r = O_A(r^{-A})$ for every $A > 0$. On compact α -sets the generalized binomial coefficient grows at most polynomially in r , so the series is normally convergent. $\square$

For $\alpha = p \in \mathbb{N}_0$, the series terminates and reduces to (10). For a negative integer it remains infinite but rapidly convergent. There is no singular transition at $\alpha = -1, -2, \dots$; the flat Fabius endpoint removes the Mellin poles that an ordinary nonzero density would have.

9.2 Universal endpoint expansion

Theorem 9.3 (Complex-power Euler–Maclaurin expansion). *For every compact $K \subset \mathbb{C}$ and every fixed $R \geq 1$, uniformly for $\alpha \in K$,*

$$S_m(\alpha) = I(\alpha) + \frac{h}{2} + \sum_{q=1}^{R-1} \frac{B_{2q}}{(2q)!} \alpha^{2q-1} h^{2q} + O_{K,R}(h^{2R}) \quad (m \rightarrow \infty). \quad (46)$$

Proof. The function $x^\alpha F(x)$ extends smoothly and flatly through $x = 0$ for all α in a compact set. At $x = 1$, every positive derivative of F vanishes and $F(1) = 1$, hence

$$\left. \frac{d^r}{dx^r} (x^\alpha F(x)) \right|_{x=1} = \alpha^r.$$

Apply Euler–Maclaurin to the right rule. All lower-end corrections vanish. $\square$

For natural p , the falling factorial becomes zero beyond degree p ; at a dyadic level above the alias threshold, [theorem 6.2](#) upgrades this truncated asymptotic formula to an exact equality. For nonintegral α the series does not truncate. Subtracting $h/2$ and then the h^2 term exposes the expected even-power convergence, as shown in ????

For a fixed negative power $\alpha = -s$, the first grid point is not dangerous:

$$h^{1-s} F(h) = 2^{m(s-1)} \frac{2^{-m(m+1)/2}}{m!} P_m(1), \quad (47)$$

which decreases faster than every geometric sequence in m for fixed s . This gives a concrete discrete counterpart of Mellin entire-ness.

10 Iterated sums of arbitrary integer order

Let $a_k = (kh)^p F(kh)$ and define the inclusive discrete integral

$$(\mathcal{J}_h a)_K = h \sum_{j=0}^K a_j, \quad \mathcal{J}_h^n = \underbrace{\mathcal{J}_h \circ \cdots \circ \mathcal{J}_h}_{n \text{ times}}.$$

Theorem 10.1 (Hockey-stick kernel). *For $n \geq 1$,*

$$(\mathcal{J}_h^n a)_K = h^n \sum_{k=0}^K \binom{K-k+n-1}{n-1} a_k. \quad (48)$$

At $K = N$ this becomes

$$(\mathcal{J}_h^n [x^p F(x)])_N = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} c_{n,j}(h) S_{m,p+j}, \quad (49)$$

where

$$\prod_{r=1}^{n-1} (1+rh-x) = \sum_{j=0}^{n-1} c_{n,j}(h) x^j, \quad c_{n,j}(h) = (-1)^j e_{n-1-j}(1+h, 1+2h, \dots, 1+(n-1)h). \quad (50)$$

Proof. Repeatedly interchange the finite sums; the number of weak chains between a fixed k and K is the binomial coefficient in (48). At $K = N$,

$$h^n \binom{N-k+n-1}{n-1} = \frac{h}{(n-1)!} \prod_{r=1}^{n-1} (1+rh-kh).$$

Expand the polynomial and identify the ordinary comb moments. □

This answers the n th-order question without introducing a new family of unknown constants. Combining [theorems 6.2](#) and [10.1](#) gives the explicit stabilized rule

$$\begin{aligned} (\mathcal{J}_h^n [x^p F(x)])_N &= \frac{1}{(n-1)!} \sum_{j=0}^{n-1} c_{n,j}(h) \left[I_{p+j} + \frac{h}{2} \right. \\ &\quad \left. + \sum_{q=1}^{\lfloor (p+j+1)/2 \rfloor} \frac{B_{2q}}{(2q)!} p + j^{2q-1} h^{2q} \right], \end{aligned} \quad (51)$$

valid, conservatively, for $m \geq p+n$. It is an exact polynomial in h with rational coefficients.

The continuum limit is the usual n -fold integral:

$$\lim_{m \rightarrow \infty} (\mathcal{J}_h^n [x^p F(x)])_N = \frac{1}{(n-1)!} \int_0^1 (1-x)^{n-1} x^p F(x) dx = \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} I_{p+j}. \quad (52)$$

For integral p, n , this limit is rational by [theorem 3.1](#).

A strict nesting convention, in which every outer index is larger rather than larger-or-equal, replaces the kernel by $\binom{K-k}{n-1}$ and changes $1+rh-x$ to $1-rh-x$ (after the corresponding endpoint adjustment). The same polynomial reduction applies.

10.1 Fractional-order repeated sums

For $\Re \nu > 0$ the standard discrete Riemann–Liouville continuation is

$$(\mathcal{J}_h^\nu a)_N = h^\nu \sum_{k=0}^N \frac{\Gamma(N-k+\nu)}{\Gamma(\nu)\Gamma(N-k+1)} a_k. \quad (53)$$

At $\nu = n \in \mathbb{N}$ it agrees with (48). Combining it with (44) gives a finite exact expression for arbitrary complex power α and fractional order ν . Under ordinary compactness conditions its continuum limit is

$$\frac{1}{\Gamma(\nu)} \int_0^1 (1-x)^{\nu-1} x^\alpha F(x) dx. \quad (54)$$

Unlike the integer-order kernel, the fractional kernel is not a polynomial, so one should not expect a finite reduction to ordinary moments; a Newton-series reduction is still available.

11 Exact and numerical checks

All exact tables in this section are generated by the accompanying Python program from rational Bernoulli numbers, cumulants, Bell recurrences, and the sparse Thue–Morse differences. No floating-point values enter the proofs. The fractional-power table uses 90–100 decimal digits and a level-15 endpoint-corrected reference.

ed.: the generated numerical results file (`generated_results.tex`) was not shipped in the archive; the tables and closed forms it would insert here are unrecoverable. The companion package `Fabius_Dyadic_Comb_Sums_Report_Package/` in this group carries its own `generated/` data tables for the same circle of identities.

12 Further deductions

12.1 A quadrature-design interpretation

The exactness in [theorem 6.1](#) is a Strang–Fix-type phenomenon in an unusual guise. Ordinarily, polynomial reproduction is encoded by zeros of a Fourier transform at a reciprocal lattice. Here the reciprocal-lattice zeros are not merely present: their multiplicity increases linearly with the dyadic level because each refinement introduces another sinc factor that vanishes at all aliases. The degree of exactness therefore grows with refinement while the node weights remain the samples $hu(kh)$ themselves.

For the Fabius half-bump, cutting at the center removes one zero through the factor $(e^{iz} - 1)^{-1}$. The only residual aliases below degree m are the elementary endpoint pole, whose zeta sums are exactly the Bernoulli corrections. This gives a conceptual reason why the finite formula (29) looks like Euler–Maclaurin but is stronger than it.

12.2 Two-adic filtration of higher defects

For $p \geq m$, write $\ell = 2^r q$ with q odd. The local zero order of R is $m+r$, so only $0 \leq r \leq p-m$ occur. Hence

$$\mathcal{R}_{m,p} = \sum_{r=0}^{p-m} \mathcal{R}_{m,p}^{[r]},$$

where each stratum is an odd-integer Dirichlet series built from a finite jet of the zero-stripped sinc product at πq . The $r=0$, $p=m$ jet is exactly $D(m+1)$. Higher jets involve $H^{(j)}(\pi q)$ and finite cotangent sums. This filtration is finite for every fixed pair (m, p) and should be preferable to directly differentiating the whole infinite product.

12.3 Richardson extrapolation

For complex α , define $\tilde{S}_m(\alpha) = S_m(\alpha) - 2^{-m-1}$. Then (46) contains only even powers 2^{-2m} . Standard Richardson or Romberg elimination therefore applies with ratio 4. When $\alpha = p \in \mathbb{N}_0$, the formal correction sequence is finite; once $m \geq p + 1$, eliminating all nonzero powers returns I_p exactly in rational arithmetic. For nonintegral α , the same scheme gives rapidly improving approximants to the entire Mellin moment $I(\alpha)$.

13 Conjectures

The following claims are intentionally separated from the proved results. They are supported by the exact experiments bundled with the report and by the local alias structure, but a complete proof was not obtained here.

Conjecture 13.1 (Sharp first nonzero defect). *For every $m \geq 1$, the smallest p for which $\mathcal{R}_{m,p} \neq 0$ is*

$$p = \begin{cases} m, & m \text{ odd}, \\ m + 1, & m \text{ even}. \end{cases}$$

Equivalently, apart from the forced even boundary cancellation in theorem 6.2, there are no accidental cancellations among the odd-alias jets.

Conjecture 13.2 (Spectral positivity). *The rational numbers*

$$d_r := \frac{D(2r)}{\pi^{2r}}$$

are positive for every $r \geq 1$.

The summands in $D(2r)$ are not all positive, so this is not termwise trivial. The $q = 1$ term dominates for large r ; the difficult part is a uniform estimate covering the small exponents.

Conjecture 13.3 (Arithmetic closure of higher alias jets). *After grouping by 2-adic valuation and by Fourier parity, every parity-compatible higher-jet series needed for an integer comb defect reduces to a rational multiple of the appropriate power of π . More concretely, the combinations of*

$$\sum_{q \text{ odd}} \frac{H^{(j)}(\pi q)}{q^s}$$

that occur in $\mathcal{R}_{m,p}$ belong to $\mathbb{Q}\pi^{s-j}$ whenever the total parity makes the comb sum real.

Rationality of the *total* defect follows from the finite formula; the content of the conjecture is that the valuation/jet decomposition can be made arithmetically diagonal rather than relying on cancellation between distinct series.

14 Promising research directions

Closed recurrences for $D(s)$. Use $H(z) = \text{sinc}(z/2)H(z/2)$ to split odd residues modulo 2^r and seek a Mahler-type functional equation for the entire function $D(s)$. Exact values at positive even integers provide unusually rigid interpolation data.

Explicit higher-defect algebra. Develop the zero-stripped Taylor coefficients in (35) through Bell polynomials of logarithmic derivatives. Grouping by $v_2(\ell)$ should produce a finite symbolic algorithm for $\mathcal{R}_{m,p}$ that is substantially smaller than the direct dyadic-value sum when $p - m$ is small.

q -binomial compression. The polynomial $\Theta_m(z)$ is the finite Thue–Morse product, while P_m is controlled by the dyadic product $M(t)$. A mixed generating function in m , z , and the moment index may reveal a direct bridge to the q -binomial identities already developed in the repository, potentially compressing (20) further.

Uniform complex- α error bounds. Combine the endpoint expansion (46) with the known Lambert- W asymptotics of F near zero to obtain optimally truncated bounds uniform in vertical strips of α . The key technical question is how the flat but nonanalytic endpoint controls the beyond-all-orders remainder.

Fractional discrete calculus. Study the two-parameter function in (53) jointly in (α, ν) . Its poles and zeros under analytic continuation in ν may encode a discrete analogue of beta-moment identities for the inverse Fabius function.

Multivariate cubature. Tensor products of theorem 6.1 immediately give compactly supported exact cubatures. Non-tensor dyadic refinements, anisotropic sinc products, and sparse grids are natural next steps.

Formal verification. The rational moment recurrence, sparse-difference generation, Prouhet cancellation, and finite iterated-sum reduction are well suited to Lean. The analytic core can be split into reusable lemmas about infinite products, Poisson summation, zero multiplicity, and removable singularities.

15 Conclusion

The weighted dyadic comb is much more rigid than a generic Riemann sum. At an arbitrary level, theorems 5.1 and 5.2 give a finite rational answer in terms of Thue–Morse signs and Appell moment polynomials. Above a simple degree threshold, Poisson summation collapses that answer to a continuous moment plus finitely many universal Bernoulli corrections. The mechanism is the growing multiplicity of the sinc-product zeros at dyadic aliases.

That same mechanism yields exact spectral Dirichlet values and a finite 2-adic filtration of every higher defect. Flatness at the Fabius endpoint makes negative and fractional powers analytically benign, while the integer case acquires exact finite termination. Finally, repeated summation introduces no essentially new integer-order constants: its binomial kernel reduces it to ordinary comb moments, and its limit is a rational beta-weighted moment.

A Recursive repository inventory

The table below was generated from every `.tex` file below the requested repository path, including all subdirectories. Tags are lexical aids for the audit, not semantic classifications. The companion `corpus_audit.txt` records file sizes, first parsed headings, keyword hit counts, the commit hash, and a combined SHA-256 digest.

ed.: the relative-path corpus inventory (`corpus_inventory.tex`) was not shipped in the archive; the audited file listing this appendix would reproduce is unrecoverable.

B Reproducibility notes

Run


```
python experiments.py \
  --repo-docs /path/to/ProveIt/Analysis/FabiusFunction/docs \
  --out .
pdflatex fabius-dyadic-comb-sums.tex
pdflatex fabius-dyadic-comb-sums.tex
```

The program uses `exact fractions.Fraction` arithmetic for every integer-power theorem and table. High-precision fractional-power experiments use `mpmath`; the optional figure uses `matplotlib`. If those optional packages are absent, the exact outputs are still generated.

References

- [1] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174.
- [2] V. L. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Mathematical Surveys* **45**:1 (1990), 87–120.
- [3] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics*, 2nd ed., Addison–Wesley, 1994.
- [4] E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton University Press, 2003.
- [5] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction documentation*, recursive source corpus at commit `[not shipped]`, audited in this report.